Name_____ Date_____ Period_____

LESSON 4-3 · HOMEWORK PACKET · LEVEL 1 · SHEET 1 OF 2

Estimate with Benchmarks

Maryland Standard 6.AT.4 · Learning target – I can estimate a percent by comparing an amount to benchmark percents.

REMEMBER

- **Benchmarks:** 10%, 25%, 50%, 75%, 100%. Build others from 10%: 20% = 2 × 10%, 30% = 3 × 10%.
- Find benchmark amounts of the **whole**. Which one is the part **about** equal to?
- A **double number line** lines up amounts with percents.

WORKED EXAMPLE

A water jug holds 40 cups. There are 29 cups left. About what percent of the jug is left?

Step 1 Benchmarks of 40: 50% = 20 and 75% = 30.

Step 2 29 is very close to 30.

Step 3 So 29 out of 40 is about 75%.

Answer **about 75%**

WORD BANK

benchmark (porcentaje de referencia) – an easy percent: 10%, 25%, 50%, 75%

the whole (el entero) – the full amount; it is 100%

about (aproximadamente) – close to, not exact

estimate (estimar) – a close answer

PRACTICE

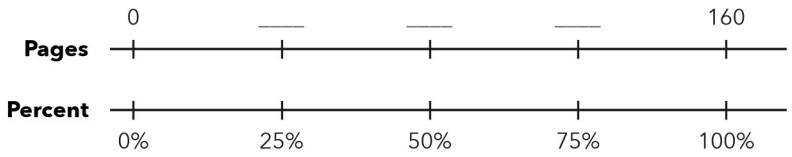
show your work · label units

1. Reading Goal. Jada's goal is to read 160 pages. She has read 42 pages. Fill in the double number line. Then estimate the percent of her goal she has read.

Step 1 Find 160 ÷ 4. That is one jump on the top line.

Step 2 Fill in each jump on the top line.

Step 3 Find the mark closest to 42.



About_____ %

2. About what percent? Name the closest benchmark: 10%, 25%, 50%, 75%, or 100%.

Step 1 Find half of the whole. Then find one fourth and one tenth.

Step 2 Which amount is the part closest to? Name that percent.

a. 7 out of 60 ≈ ___ %

b. 38 out of 50 ≈ ___ %

c. 23 out of 90 ≈ ___ %

d. 19 out of 40 ≈ ___ %

e. 49 out of 50 ≈ ___ %

f. 11 out of 45 ≈ ___ %

g. 16 out of 150 ≈ ___ %

h. 61 out of 80 ≈ ___ %

3. Class Fundraiser. A class wants $500. They have **raised** $96 so far. Use 10% of $500 to estimate the percent they have raised.

Step 1 Find 10%: $500 ÷ 10.

Step 2 Double it to get 20%. Is $96 close?

a. 10% of $500 = _____

b. About what percent have they raised? _____ %

4. Bus Seats. A bus has 48 seats. 35 seats are filled. About what percent of the seats are filled? Name the benchmark you used.

Step 1 Find 50% and 75% of 48.

Step 2 Which amount is 35 closest to?

About_____ %

5. Compare. Which is closer to 50%: 26 out of 50, or 26 out of 60? Explain how you know.

Frame ___ *out of* ___ *is closer to 50% because half of 50 is* ___ *and half of 60 is* ___.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Look at a battery icon at home. Estimate its percent and name your benchmark.

Name_____ Date_____ Period_____

LESSON 4-3 · HOMEWORK PACKET · LEVEL 1 · SHEET 2 OF 2

Estimate a Percent of a Number

Maryland Standard 6.AT.4 · Learning target – I can estimate a percent of a number using friendly percents and rounded amounts.

REMEMBER

- Round to **friendly numbers** first.
- 10%: divide by 10. 5% = half of 10%.
- 25% = 1/4, 50% = 1/2, 75% = 3/4.

WORKED EXAMPLE

Estimate 24% of 83.

Step 1 Round: 25% of 80.

Step 2 25% = 1/4, and 80 ÷ 4 = 20.

Answer **about 20**

WORD BANK

estimate (estimación) – a close answer, not exact

round (redondear) – change to a friendly number

10% (diez por ciento) – the whole ÷ 10

compatible (friendly) numbers (números compatibles) – easy to use, like 40 or 200

PRACTICE

show your work · label units

1. Estimate. Round to friendly numbers first.

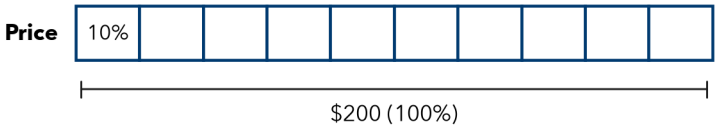
Step 1 Round the percent and the number.

Step 2 Find 10%, then build the percent you need.

a. 11% of 59 ≈ ___ **b.** 74% of 41 ≈ ___ **c.** 26% of 203 ≈ ___

2. **Sneaker Sale.** Sneakers cost $198 and are 30% off. Round to $200. Each part of the tape is 10% of $200.

Frame *One part is $200 ÷ ___ = ___. The discount is 3 × ___ = ___.*



a. About how many dollars is one part (10%)? _____

b. About how much is the discount (30%)? _____

3. **School Lunch.** 412 students. About 48% **buy lunch**. About how many students buy lunch?

Step 1 Round: 48% is about 50%. 412 is about 400.

Step 2 50% is half. Find half of 400.

About_____ students

4. Apple Orchard. A farm picked 798 apples. About 24% of them were too small to sell. About how many apples were too small?

Step 1 Round: 24% is about 25%. 798 is about 800.

Step 2 25% is one fourth. Divide by 4.

About_____ apples

5. Find the Error. Mia estimated 19% of 512 by writing 20 × 500 = 10,000. Explain her mistake. Give a good estimate.

Frame *Mia's mistake is ___. 20% means ___, so a good estimate is ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. Which is the best estimate of 26% of 79?

Step 1 Round: 26% is about 25%. 79 is about 80.

Step 2 25% is one fourth. Divide by 4.

A 2 **B** 20 **C** 26 **D** 40

7. A coat costs $119. It is 40% off. Which is the best estimate of the discount?

Step 1 Round $119 to $120.

Step 2 Find 10% of $120, then multiply by 4.

A $4.80 **B** $40 **C** $48 **D** $72

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Find a sale sign. Estimate the discount out loud with someone at home.